

# ACFI315 Principles of finance with Excel

## Week 5 Mean-Variance Analysis

Two data files have been uploaded to VITAL.

### 1. Apple and Tesla Data

The file contains daily data on the price of equity in Apple and Tesla sampled over the period from 21<sup>st</sup> October 2019 to 18<sup>th</sup> October 2021. The expected return and variance of return on a two-asset portfolio are given by

$$E(\bar{R}_p) = aE(\bar{R}_X) + bE(\bar{R}_Y)$$
$$VAR(\bar{R}_p) = a^2VAR(\bar{R}_X) + b^2VAR(\bar{R}_Y) + 2abCOV(\bar{R}_X, \bar{R}_Y)$$

Where  $a$  is the proportion of wealth invested in asset  $X$  and  $b = 1 - a$ . Given that  $\rho_{XY}\sigma_X\sigma_Y = COV(X, Y)$ , we may write the variance of the portfolio as

$$VAR(\bar{R}_p) = a^2VAR(\bar{R}_X) + b^2VAR(\bar{R}_Y) + 2ab\rho_{XY}\sigma_X\sigma_Y$$

An analytic solution for the variance minimizing weight can be found using

$$a^* = \frac{\sigma_Y^2 - \rho_{XY}\sigma_X\sigma_Y}{\sigma_X^2 + \sigma_Y^2 - 2\rho_{XY}\sigma_X\sigma_Y} \text{ where } \rho_{XY}\sigma_X\sigma_Y = COV(X, Y)$$

Use the data on Apple and Tesla to complete the following tasks:

- Calculate weekly log returns on the two stocks using the formula  $r_t = \ln\left(\frac{P_t}{P_{t-1}}\right)$ .
- Find the average return and the standard deviation of returns of each stock.
- Calculate the coefficient of correlation between the two returns series ( $\rho_{XY}$ ).
- Create a scatter plot showing the relationship between the returns on the two stocks. Fit a linear trend line to the plot and verify that  $\rho_{XY}^2 = R^2$  of the fitted linear regression.
- Using portfolio weights in increments of 0.05, calculate the expected return and risk (volatility) of a portfolio comprising the two stocks.
- Plot the risk-return frontier of the portfolio. What happens to the shape of the frontier when  $\rho_{XY} = 1$  or  $-1$ .
- Find the asset weights associated with the minimum variance portfolio analytically.
- Verify your results in (g) using Solver.

### 2. SP500 Daily

The file contains daily data on the Standard and Poor's 500 index sampled on a daily basis from 23<sup>rd</sup> October 2017 to 21<sup>st</sup> October 2022. Use the data to complete the following questions:

- Calculate the continuously compounded returns on the index using the formula  $r_t = \ln\left(\frac{P_t}{P_{t-1}}\right)$ .
- Graph the continuously compounded returns against time. What pattern is apparent in the volatility of the return series?
- Calculate rolling 5-day standard deviations of the continuously compounded returns you created in (a).
- Add a line plot of the rolling 5-day standard deviation of the plot created in (b). Make sure to include a secondary y-axis in your graph.
- Use the PERCENTILE.INC function to find the historical daily 99% Value-at-Risk of S&P500.